1. Short answer type questions : 2×7=14

(a) How are the elements of array accessed ?

(b) Explain the purpose of setwd().

(c) Write a program to find the minimum value in a vector.

(d) Explain the purpose of is() used in R.

(e) What do you mean by data melting() ?

(f) What do you mean by vector recycling ?

(g) How can you import the data from Excel file in R ?

Unit I

2. What are the features of R ? Write down the advantages of R language over other programming languages of this category. 14

3. Write and explain different types of data types supported by R-programming language. 14

Unit II

4. (a) Differentiate between while loop and for loop.

(b) Illustrate the different logical operators supported in R-Programming language.

7,7

5. Write the R code for the following : 14

(a) Calling a function with arguments

(b) Calling a function without arguments.

Unit III

6. (a) Create a matrix and write a R code to Generate the following :

(i) Access the element at 3rd column and 1st row in a matrix.

(ii) Access only the second row

(iii) Access the element at 2nd column and 3rd row in a matrix.

(b) Illustrate the manipulation of String data using : paste(), grep() and toupper(). 7,7

7. (a) Create a data frame with at least eight rows and apply head() and tail().

(b) How can we merge the two data frames ? Explain with a R program. 7,7

Unit IV

8. What is a CSV File ? How can you read a CSV file in R ? Illustrate min(), max() and mean() on CSV file in R. 14

9. Illustrate the plotting of Histogram with syntax and example. Explain briefly about the ggplot2 package. 14